

# BMD101 Integration Reference Design/Schematic

## Reference Schematic Design

BMD101™ has a single power input, VDD. To limit the noise that is injected into the ECG signal, we recommend either:

- Pi Filter, as shown in Figure 1.1
- DC-DC Regulator, which can output 3.3V

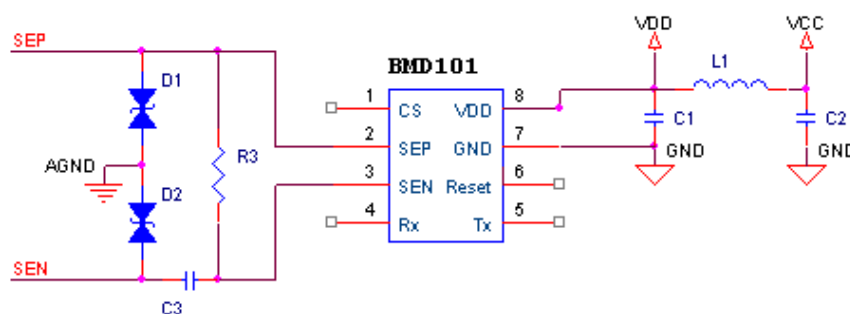


Figure 1: Reference Schematic for ECG Applications

| Reference Component | Recommend Value                                      | Note                                 |
|---------------------|------------------------------------------------------|--------------------------------------|
| L1                  | 10 uH                                                | Pi Filter component                  |
| C1,C2               | 10 uF                                                | Pi Filter components                 |
| D1,D2               | For Toys 4kV Contact Discharge and 8kV Air Discharge | ESD Diodes                           |
| R3,C3               | See "External High Pass Configuration" Section       | External High Pass Filter (Optional) |

## Isolated Digital and Analog Ground

To reduce noise in the ECG signal, isolate Analog and Digital Grounds for BMD101. SEP/SEN are best connected to Analog Ground, isolating the sensitive sensor lines from potential power supply noise.

For the BMD101 Starter Kit Module, a 10uH inductor is used to isolate the separate Grounds.

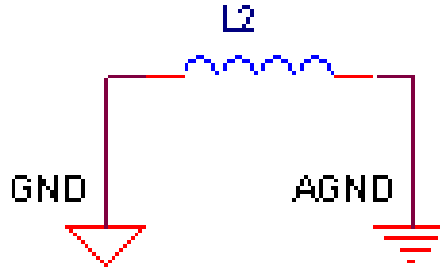


Figure 2: Ground Isolation Schematic

## Shielded Connections for SEP/SEN

For best results, shielded wiring is recommended for the SEP/SEN sensor connections.

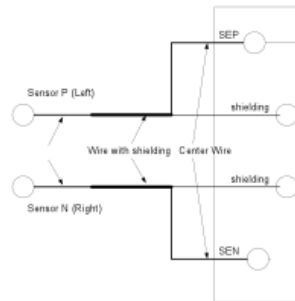


Figure 3: Shielding Connections

## External High Pass Configuration (Optional)

BMD101 is equipped with an internal high pass filter. For certain high-noise situations, such as running or bicycling, additional filtering may be preferred. The following section will outline how to use an external high pass filter in conjunction with BMD101.

Please refer to Figure 1. Please note the HPF (C3,R3) at the SEP/SEN inputs of BMD101. We can calculate the cutoff frequency by using,

$$f_c = \frac{1}{2\pi\tau} = \frac{1}{2\pi RC}$$

Figure 4: High Pass Filter Cutoff Frequency Equation

If you choose not to use a HPF, populate C3 with a 0 ohm resistor (0603).

## Reference PCB Layout

An example of the split ground planes for Analog and Digital Ground is shown in Figure 2.1 below. Note that the planes split **underneath** BMD101.

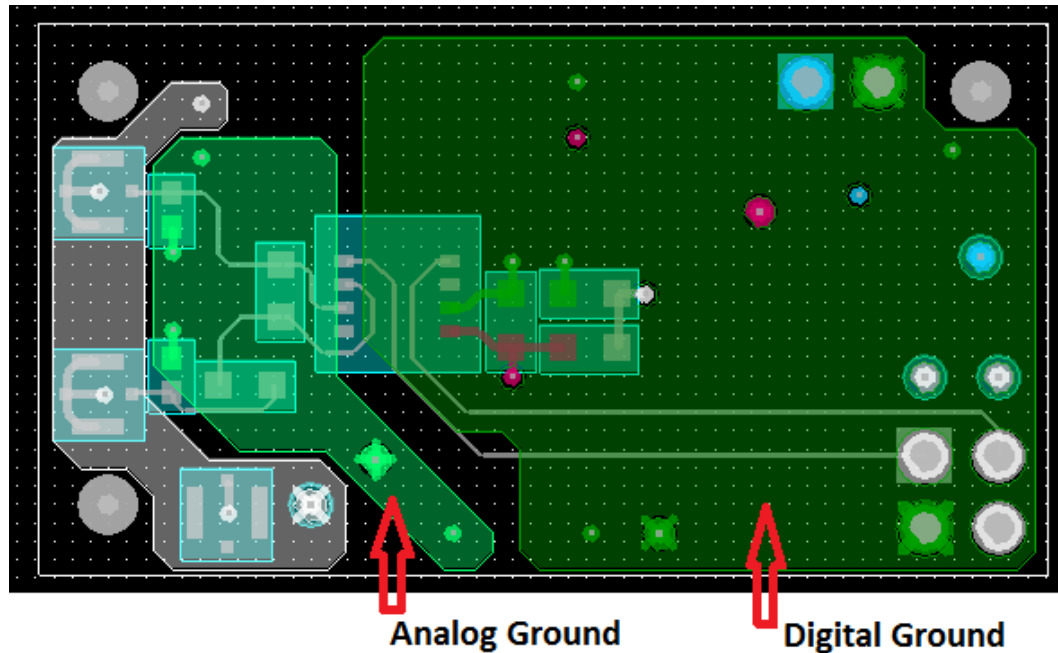


Figure 5: Split Ground Layout

## Sensor Specification

Testing was conducted at NeuroSky to rate the performance of various sensor materials and sizes. The results and recommended sensor material and size are listed in the following sections.

### Testing Results

BMD101 will produce an ECG signal if the subject merely holds a wire connected to the sense leads. Any sensor material will produce an accurate QRS wave. What sets the different materials apart is the clarity of P & T waves.

The QRS waveform was consistent among all sensor materials. The height and width varied from subject to subject, but amongst a single subject, the measurements were uniform.

**Silver-Silver Chloride** exhibited the strongest signal quality. It had the best signal to noise ratio, which increased the clarity of both P and T waves. It also had the least amount of low frequency drift.

**Stainless steel** and **Gold** scored similar grades. Both noisier than silver-silver-chloride, their signals were not as clean, masking the presence of P waves in some subjects.

**Conductive rubber** scored strongly as well. It was on similar to gold and stainless steel. QRS was consistent with the other materials. P and T weren't as prominent, but the sensor wasn't ideal for finger testing. When placed on the chest, the signal is very clean and stable. Please note that different grades of conductive rubber will behavior differently.

## Sensor Specifications for ECG Applications

| Material               | Recommended Diameter |
|------------------------|----------------------|
| Stainless Steel        | 10 mm                |
| Silver Silver Chloride | 8 mm                 |
| Gold                   | 10 mm                |
| Conductive Rubber      | 12 mm                |

**Note: If using non-circular sensors, substitute length/width for diameter.**

**Corporate Address**

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